Guidelines for the exam in “Advanced Signal Processing”

The exam is oral with approximately 15 minutes allocated for examining each student. In addition, for each student, approximately 5 minutes will be used for evaluation among the examiners.

The course consists of 3 areas, each containing multiple topics. Accordingly, the examined areas will be:

Area A topics:
1. Adaptive filtering: Least Mean Squares (LMS)
2. Adaptive filtering: Recursive Least Squares (RLS)
3. Multi-dimensional Fourier transform

Area B topics:
4. Detection theory
5. Minimum Mean Squared Error (MMSE) estimation
6. Kalman Filtering (scalar and vector versions)

Area C topics:
7. Linear predictive coding: The General overview
8. Linear predictive coding: Filter design by modeling
9. Liner predictive coding: The lattice filter

The student will be examined in two topics belonging to two different areas. At the beginning of the exam, the student draws randomly two topics – Topic 1 and Topic 2, which belong to two different areas, A, B, or C, as listed above. The student decides the order in which the topics should be presented and discussed.

The examination will consist of a discussion during which the student initially presents his/her knowledge/understanding of the topics, using the black board for figures, equations, etc. Furthermore, the student will also be asked multiple specific clarifying questions within the two given topics. The examiners direct the questions based upon the answers of the student. The questions will address both the conceptual and mathematical understanding of the selected topics. As a rule of thumb, the examiners will aim to spend the same amount of time in each topic. Students are allowed to bring to the exam one sheet of notes per topic (i.e., 9 in total) with the contents that they consider necessary to present the topics.

The course is assessed on the 7-step grading scale. The examination will be conducted in English/Danish, and the examiners will be the course lecturers, i.e., Jesper Rindom Jensen, Troels Pedersen, and Peter Koch

Aalborg University, November 21, 2023

Jesper, Troels, and Peter